

Alex Mercer

SOFTWARE ENGINEER

alex.mercer@email.com | +1 (555) 019-2834 | github.com/alexmercerc | linkedin.com/in/alexmercerc

EDUCATION

Tech University

Graduation: May 2025

Bachelor of Science in Computer Science

- Key Coursework: Advanced Data Structures, Database Management Systems, Operating Systems, Software Engineering Principles.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript (TypeScript), HTML/CSS, SQL

Frameworks & Tools: React.js, Node.js, Express, FastAPI, Tailwind CSS, Linux/Bash, Git/GitHub, Docker, AWS (S3, EC2)

Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), RESTful APIs, System Design

EXPERIENCE

Software Engineering Intern

June 2024 – August 2024

Nexus Tech Solutions

- Collaborated in an Agile team to migrate a legacy monolithic backend to a microservices architecture using Node.js and Docker, reducing server response times by 15%.
- Optimized database queries in PostgreSQL, which decreased API latency by 25% for high-traffic endpoints.
- Wrote over 50 automated unit tests using Jest, increasing overall code coverage from 72% to 88%.

PROJECTS

Distributed Task Queue

C++, Linux, Threads

- Built a high-performance, multi-threaded task queue in C++ to handle asynchronous job execution.
- Implemented thread pools and mutex locks to ensure thread safety, managing up to 10,000 concurrent tasks with zero race conditions.
- Designed a custom logging system to track execution metrics and debug process bottlenecks.

Cloud-Native Metrics Aggregator

Python, FastAPI, React, Docker

- Developed a full-stack telemetry monitoring tool that captures and visualizes system infrastructure parameters in real-time.
- Built a lightweight backend using FastAPI to process metrics data and expose structured REST endpoints.
- Created a responsive React frontend with interactive data streaming components to display live bandwidth usage and anomaly flags.

| ACHIEVEMENTS & CERTIFICATIONS

- **Competitive Programming:** Max rating of 1620 (Expert) on Codeforces; solved 500+ algorithmic problems across various platforms.
- **Hackathon Winner:** Secured 2nd place out of 80+ teams in a regional collegiate hackathon for developing an AI-driven automation script within 36 hours.
- **AWS Certified Cloud Practitioner** (Expired/Active status maintained since 2025)